INF10025 Learning Summary Template

Big Data Management (Swinburne University of Technology)



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LEARNING SUMMARY REPORT INF10025–Data Management and Analytics

Emmanuel Edwin (103599064)

Self-Assessment Details

The following checklists provide an overview of my self-assessment for this unit.

	Pass (P)	Credit (C)	Distinction (D)	High Distinction (HD)
Self-Assessment (please tick)		✓		

Self-assessment Statement

	Included (please tick)
Learning Summary Report _ Part 1 Completed	✓
Test 1 and Test 2 are Completed (at least 10/20)	✓
At least 4 Tasks are Completed	✓

Minimum Pass Checklist

	Included (please tick)
Learning Summary Report _ Part 1 Completed	✓
Test 1 and Test 2 are Completed (at least 10/20)	✓
At least 8 Tasks Completed	✓

Minimum Credit Checklist

	Included (please tick)
Learning Summary Parts 1 and 2	
All 5 Pass Tasks Completed	
All 5 Credit Tasks Completed	
D task is at Satisfactory Level	
Test 1 and Test 2 are Complete (at least 10/20 on first attempt)	
Test 3 is Complete	

Minimum Distinction Checklist

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	Included (please tick)
Learning Summary Parts 1, 2 & 3 Completed	
All 5 Pass Tasks Completed	
All 5 Credit Tasks Completed	
D task is at Satisfactory Level	
HD task is at Satisfactory Level	
Test 1 and Test 2 are Complete (at least 10/20 on first attempt)	
Test 3 is Complete	

Minimum High Distinction Checklist

Comments

This is the section to provide comments on how your work meets the relevant grade requirements.

I am aiming for a 64 but I have also met all the requirements for a Credit (60) with 5/5 Pass Tasks and 3/5 credit tasks with one pending marking. I believe I have completed quality work, and I did well in my tests after failing the firs attempt, I stepped up my game. I am proud of my work, and I am certain I should achieve a Credit.

Declaration

I declare that this portfolio is my individual work. I have not copied from any other student's work or from any other source except where due acknowledgment is made explicitly in the text, nor has any part of this submission been written for me by another person.

Signature: 

If the following section does not contain reflection on your learning, **you will not get more than 50** for this unit, regardless of other submissions and achievements.

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PART 1 – All students to Complete

Introduction

This report summarises what I learnt in INF10002 Database Analysis and Design. It includes a self-assessment against the criteria described in the unit outline, a justification of the pieces included, details of the coverage of the unit's intended learning outcomes, and a reflection on my learning.

Coverage of the Intended Learning Outcomes

This section outlines how the pieces I have included demonstrate the depth of my understanding in relation to each of the unit's intended learning outcomes.

Reflections on Access and PowerBI

ILO 1:

Define and explain fundamental data and database concepts including tables, relations, keys, queries, transactions, and structured, semi-structured, unstructured data

The following pieces demonstrate my ability in relation to this ILO:

- Tasks 1 P and C

  My first 2 tasks received an 10/11 and 8/10 my learning points was cardinalities not being shown in the relationships. Task 1 really taught me how to manage large amounts of data and how to accurate pinpoint the relevant data a user may be looking for, by using basic queries in the Query design on Microsoft Access. I also learnt about basic entity and data relationships, for example how crucial the understanding of Primary Keys and Foreign keys and their impact on tables of data.

- Test 1

  Test 1 really stretched my learning, I was not prepared for it as much as I thought I was, although we covered basic queries and parameters whilst completing Task 1, I still lacked some logical thinking when faced with the query questions on the test, I ended up scoring 8/20 on my first attempt and I was disappointed to say the least. When test 1 resit 1 was approaching I did a lot more revision to make I was able to have a logical viewpoint/mindset when I read every "IF" "OR" and "AND" parameter questions and I managed to pass with 13/20 which was a reasonable step up. Almost glad I failed in the end because it forced me to strengthen my knowledge before doing the other assignments.

ILO 2:

Create, store, retrieve, exploit, and visualise data using modern database tools, functions, and techniques

- Task 2 P

I did not have time to attempt and submit Task 2 C as that week it was due I was struggling with time management with all 4 units having things due at the same time. I wish I attempted it looking back now. Task 2 P was actually a very fun assignment, data visualization is a such a useful tool when working with large amounts of data that would be hard to fully understand if not visualised. That being said, this was my first and last time receiving a "Re-work and resubmit" as I had not achieved 80% correctness with 10/14. My main issues were incorrect data labels and a few technical issues with MS Power BI, it was such a powerful tool with loads of options to display data. I also learnt more about storing and retrieving data, and I created test data of my own which was good to learn.

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ILO 3:

Understand and apply conventional data modelling techniques to solve practical database design problems.…

- Tasks 1 and 2

These two tasks taught me how to create and manage tables and queries. Microsoft Access is an easy-to-understand tool that allowed me to spend plenty of time revising all the techniques and parameters we learnt about. Once that was understood I learnt how to apply those conventional data modelling techniques and visualise large amounts of data in MS Power BI, this was favourite of mine because I can see it being super useful/relevant to the work place I may work at in the future.

Reflections on SQL and ERDs

ILO 1:

Define and explain fundamental data and database concepts including tables, relations, keys, queries, transactions, and structured, semi-structured, unstructured data

The following pieces demonstrate my ability in relation to this ILO:

- Task 3 P and C

Tasks 3 P and C, I got 8/10 for 3P and 9/10 for 3C. This was because I have worked with SQL in the past before as I used to study Computer Science. Nevertheless, I learnt plenty from the assignments, my only downfalls were incorrect criteria. ERDs were a favourite of mine too because they are a great visual tool to make sense of relationships and entities which can be difficult to understand by simply looking at a SQL statement. I learnt the fundamentals of data and keys especially, ERD's helped me truly understand the relationship between data and tables within a database. Which ultimately helped me prepare for test 2

- Test 2

I definitely came into Test 2 a lot more prepared than I did to Test 1, this was because I had spent more time completing my Tasks P and C which gave me very clear idea of ERD's which is where I scored the most marks for in my test which came back as 15/20. I was very happy to pass first go and with some good distance from just barely passing. This encouraged me to always put more time and effort into truly understanding the learning topics each Pass and Credit Task was focused on.

ILO 2:

Create, store, retrieve, exploit, and visualise data using modern database tools, functions, and techniques

- Tasks 3

These tasks were essential to my understanding of how to handle data properly, because all SQL statements had to be concise and some of them were long statements. I learnt how to create tables, edit data in tables, delete data, retrieve data and group/sort it. I learnt the importance of primary and foreign keys again.

ILO 3:

Understand and apply conventional data modelling techniques to solve practical database design problems.…

ERD's and relational schemas were also a huge focal point for these learning outcomes as they were essential in our understanding of relationships moving forward. These important topics really made sure we could create a clear visualisation of all the relationships and then also type it out as

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a relational schema which tests our knowledge of how it would apply to a database like using SQL for example.

Normalisations, Transactions, JSON

ILO 4:

Explain the impact of big data and modern data analytics tools in the design, use, and management of database systems. …

- Tasks 4 P and C + 5 P and C

Tasks 4 and 5 were the most time-consuming tasks because we spent a lot of time designing ERDs and Relational schemas, it was good to test our understand of how to visualize the relationships and database as a whole. I achieved a first time Pass for all of these tasks except 5C as I am still waiting for a mark for that task. The best part about these tasks was understanding and applying “normalisation” 1NF 2NF and 3NF, it was a super practical way to teach me how to actually split data up, increase it’s accuracy etc. translating the paragraphs into ERD’s was challenging as you did not want to miss a key piece of information but it expanded my capabilities at logically understanding a business process and that is so useful for me as a BA student

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Reflection

The most important things I learnt:

I am not just saying this, but this unit was very well structured and therefore I learnt a lot from Tasks 1 to 5. It was a manageable work load to I felt like I could actually learn from each assignment rather than rushing through them and failing. That being said, I have learnt so much about data and databases, the types of data, how to filter it and organise it, how to create huge databases and input data, how to visualize data effectively and also how to do all things data related as efficiently as possible.

I found the following topics particularly challenging:

I definitely think SQL was the hardest in the end because I knew a lot about it but it was hard at times when I would have an error with a statement and I couldn't figure out the issue, but it challenged me which is good.

I found the following topics particularly interesting:

Definitely visualising data, using MS Power BI and creating ERD's, they were time consuming but I can see myself using these tools in my professional career one day.

I feel I learnt these topics, concepts, and/or tools really well:

I definitely learnt the ERD and SQL statements the best, but I guess the MS Access tasks were super easy too because of how easy MS Access makes it for you with their query design.

I still need to work on the following areas:

I need to work on logical thinking some more, this may seem like a broad point but I need to take the time to really examine a problem using a logical thought process to determine the best and most effective solution.

I explored the following topics beyond the unit tasks:

Understanding of businesses and how they actually run from a technical perspective, because it wasn't the learning aim but I definitely learnt more about business processes and even how I can optimise them if I am able to in my future career

The things that helped me most were:

A lot of online tools where data rules are explained, but also Roberts teaching was quite good, his experience in this industry and topic were very obvious and he has clearly learnt how to explain topics in an easy to understand way, I found that when he used the whiteboard to explain things like inner and outer joins (real example from class) I saw, heard and forever understood from that point

This unit will help me in the future:

I've mentioned this multiple times in my report so far, but here: logical thinking, understanding of data and business processes, data visualization, managing large databases/data, problem solving and efficiency.

If I did this unit again, I would do the following things differently:

I would honestly go for a higher grade like a Distinction, I was not very effective with my time management, and the Tasks were not unreasonable by any means, so if I was able to do it again I would manage my time better in order to be more prepared and able to achieve a Distinction.

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Other…:

[Add any other reflections you think help you demonstrate your learning, feel free to include feedback]

- Discuss the approach that you used to solve problems and how the concepts learnt in this unit helped.
- Include discussion of how ideas/techniques/principles from this unit can be used in your further learning inside and outside university

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